2 Local existence of primitives and Cauchy's theorem in a disc

We first prove the existence of primitives in a disc as a consequence of Goursat's theorem.

Theorem 2.1 *A holomorphic function in an open disc has a primitive in that disc.*

Proof. After a translation, we may assume without loss of generality that the disc, say D, is centered at the origin. Given a point $z \in D$, consider the piecewise-smooth curve that joins 0 to z first by moving in the horizontal direction from 0 to $\tilde{z}$ where $\tilde{z} = \text{Re}(z)$, and then in the vertical direction from $\tilde{z}$ to z. We choose the orientation from 0 to z, and denote this polygonal line (which consists of at most two segments) by γ_z, as shown on Figure 3.

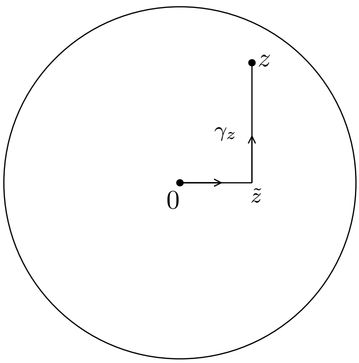

Figure 3. The polygonal line γ_z

Define

$$F(z) = \int_{\gamma_z} f(w)\, dw.$$

The choice of γ_z gives an unambiguous definition of the function $F(z)$. We contend that F is holomorphic in D and $F'(z) = f(z)$. To prove this, fix $z \in D$ and let $h \in \mathbb{C}$ be so small that $z + h$ also belongs to the disc. Now consider the difference

$$F(z+h) - F(z) = \int_{\gamma_{z+h}} f(w)\, dw - \int_{\gamma_z} f(w)\, dw.$$

The function f is first integrated along γ_{z+h} with the original orientation, and then along γ_z with the reverse orientation (because of the minus sign in front of the second integral). This corresponds to (a) in Figure 4. Since we integrate f over the line segment starting at the origin in two opposite directions, it cancels, leaving us with the contour in (b). Then, we complete the square and triangle as shown in (c), so that after an application of Goursat's theorem for triangles and rectangles we are left with the line segment from z to $z + h$ as given in (d).

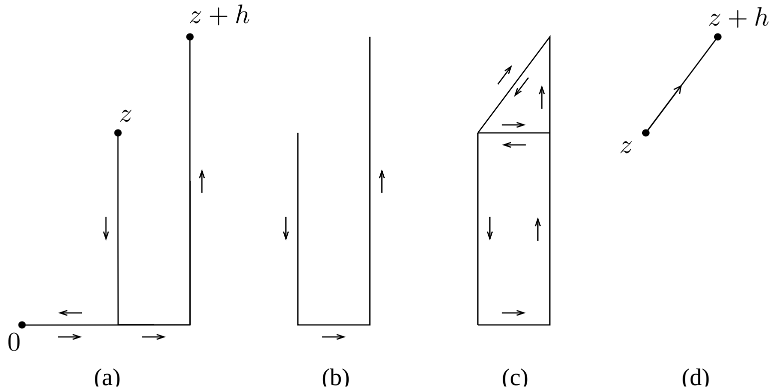

Figure 4. Relation between the polygonal lines γ_z and γ_{z+h}

Hence the above cancellations yield

$$F(z+h) - F(z) = \int_\eta f(w)\, dw$$

where η is the straight line segment from z to $z + h$. Since f is continuous at z we can write

$$f(w) = f(z) + \psi(w)$$

where $\psi(w) \to 0$ as $w \to z$. Therefore

(4)

$$F(z+h) - F(z) = \int_\eta f(z)\, dw + \int_\eta \psi(w)\, dw = f(z) \int_\eta dw + \int_\eta \psi(w)\, dw.$$

On the one hand, the constant 1 has w as a primitive, so the first integral is simply h by an application of Theorem 3.2 in Chapter 1. On the other hand, we have the following estimate:

$$\left| \int_\eta \psi(w)\, dw \right| \leq \sup_{w \in \eta} |\psi(w)|\, |h|.$$

Since the supremum above goes to 0 as h tends to 0, we conclude from equation (4) that

$$\lim_{h \to 0} \frac{F(z+h) - F(z)}{h} = f(z)\,,$$

thereby proving that F is a primitive for f on the disc.

This theorem says that locally, every holomorphic function has a primitive. It is crucial to realize, however, that the theorem is true not only for arbitrary discs, but also for other sets as well. We shall return to this point shortly in our discussion of "toy contours."

Theorem 2.2 (Cauchy's theorem for a disc) *If f is holomorphic in a disc, then*

$$\int_\gamma f(z)\, dz = 0$$

for any closed curve γ in that disc.

Proof. Since f has a primitive, we can apply Corollary 3.3 of Chapter 1.

Corollary 2.3 *Suppose f is holomorphic in an open set containing the circle C and its interior. Then*

$$\int_C f(z)\, dz = 0.$$

Proof. Let D be the disc with boundary circle C. Then there exists a slightly larger disc D' which contains D and so that f is holomorphic on D'. We may now apply Cauchy's theorem in D' to conclude that $\int_C f(z)\, dz = 0$.

In fact, the proofs of the theorem and its corollary apply whenever we can define without ambiguity the "interior" of a contour, and construct appropriate polygonal paths in an open neighborhood of that contour and its interior. In the case of the circle, whose interior is the disc, there was no problem since the geometry of the disc made it simple to travel horizontally and vertically inside it.

The following definition is loosely stated, although its applications will be clear and unambiguous. We call a **toy contour** any closed curve where the notion of interior is obvious, and a construction similar to that in Theorem 2.1 is possible in a neighborhood of the curve and its interior. Its positive orientation is that for which the interior is to the left as we travel along the toy contour. This is consistent with the definition of the positive orientation of a circle. For example, circles, triangles, and rectangles are toy contours, since in each case we can modify (and actually copy) the argument given previously.

Another important example of a toy contour is the "keyhole" Γ (illustrated in Figure 5), which we shall put to use in the proof of the Cauchy integral formula. It consists of two almost complete circles, one large

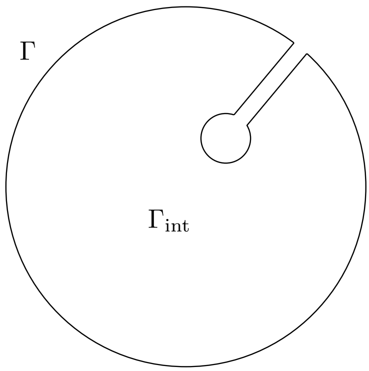

Figure 5. The keyhole contour

and one small, connected by a narrow corridor. The interior of Γ, which we denote by Γ_{int}, is clearly that region enclosed by the curve, and can be given precise meaning with enough work. We fix a point z_0 in that interior. If f is holomorphic in a neighborhood of Γ and its interior, then it is holomorphic in the inside of a slightly larger keyhole, say Λ, whose interior Λ_{int} contains $\Gamma \cup \Gamma_{\text{int}}$. If $z \in \Lambda_{\text{int}}$, let γ_z denote any curve contained inside Λ_{int} connecting z_0 to z, and which consists of finitely many horizontal or vertical segments (as in Figure 6). If η_z is any other such curve, the rectangle version of Goursat's theorem (Corollary 1.2) implies that

$$\int_{\gamma_z} f(w)\, dw = \int_{\eta_z} f(w)\, dw\,,$$

and we may therefore define F unambiguously in Λ_{int}.

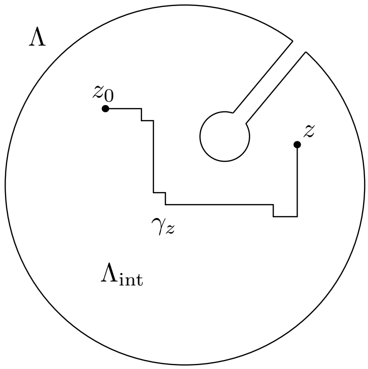

Figure 6. A curve γ_z

Arguing as above allows us to show that F is a primitive of f in Λ_{int} and therefore $\int_\Gamma f(z)\,dz = 0$.

The important point is that for a toy contour γ we easily have that

$$\int_\gamma f(z)\,dz = 0\,,$$

whenever f is holomorphic in an open set that contains the contour γ and its interior.

Other examples of toy contours which we shall encounter in applications and for which Cauchy's theorem and its corollary also hold are given in Figure 7.

While Cauchy's theorem for toy contours is sufficient for most applications we deal with, the question still remains as to what happens for more general curves. We take up this matter in Appendix B, where we prove Jordan's theorem for piecewise-smooth curves. This theorem states that a simple closed piecewise-smooth curve has a well defined interior that is "simply connected." As a consequence, we find that even in this more general situation, Cauchy's theorem holds.

3 Evaluation of some integrals

Here we take up the idea that originally motivated Cauchy. We shall show by several examples how some integrals may be evaluated by the use of his theorem. A more systematic approach, in terms of the calculus of residues, may be found in the next chapter.